



FIRST SEMESTER 2026-27
COURSE HANDOUT

Date: 30th July 2026

In addition to part I (General Handout for all courses appended to the Time table) this portion gives further specific details regarding the course.

Course No : **BITS F 429**
Course Title : **Nanotechnology for Renewable Energy and Environment**
Instructor-in-charge : **Aditya Singh**

1. Course Description:

This course provides an introduction to nanotechnology, emphasizing its applications in renewable energy and environmental engineering. It covers fundamental concepts, including the distinction between the macro- and nanoscale, nanoscale phenomena, and the distinctive properties of nanomaterials that enable them to outperform many conventional materials. The course also discusses the fundamentals of fabrication, characterization, and operating principles of advanced energy devices, including organic, quantum dot, and dye-sensitized solar cells, fuel cells, batteries, and carbon nanotube-based energy storage systems, and the application of nanomaterials for hydrogen production and storage. In addition, it examines the application of nanotechnology in improving solar cells, nanoelectrodes, nanocoating, catalyst materials for fuel cells, and battery technologies. The course further explores the role of nanotechnology in environmental sustainability, including groundwater remediation, nano-adsorbents, ecotoxicology, nanomaterial toxicity, transport and fate of nanomaterials in the environment, and the broader societal impacts of nanotechnology.

2. Scope and Objective of the Course:

Nanotechnology is widely regarded as a transformative technology with the potential to significantly benefit society across diverse fields. This course is designed to equip students with the fundamental knowledge and skills required for advanced academic research or industrial careers in nanotechnology, with particular emphasis on applications in renewable energy and environmental sustainability. The course is divided into 3 modules as described below.

- **Module 1:** General introduction to principles of nanoscience and technology
- **Module 2:** Nanotechnology applications in the renewable energy sector
- **Module 3:** Nanotechnology applications in the environmental sector.

The specific topics covered under each module are listed in the course plan (refer to section 6 of this handout).

3. Course learning outcomes

- Introduce the importance and application of nanomaterials and the origin of their unique properties.
- Summarize the fabrication methods, characterization techniques, and operating principles of solar cells, fuel cells, batteries, and CNT-based energy storage systems.
- Explain the merits and motivation in the renewable energy and environmental sectors.
- Define toxic effect of nanomaterials on human and environment, relative remediation and precautions.

4. Text Books (TB):

1. James Murday, B S Murty, P Shankar, Baldev Raj and BB Rath, "Textbook of Nanoscience and Nanotechnology" University Press – IIM ; ed 1 (2012) ISBN: 978-81-7371-738-3.
2. Javier García-Martínez, "Nanotechnology for the Energy Challenge," Wiley-VCH; ed 1 (March 22, 2010) ISBN-10: 3527324011.
3. Mark R Weisner & Jean-Yves Bottero, "Environmental nanotechnology; applications and impacts of nanomaterials", McGraw Hill, 2007, ISBN-10: 0071477500.



5. Reference Books (RB):

- L. Theodore and R. Kunz, “ Nanotechnology Environmental Implications and Solutions”, Wiley-Interscience, 2005, ISBN-10: 0471699764.
- Nanotechnology Applications for Clean Water, Edited by, N. Savage, M. Diallo, J. Duncan, A. Street, R. Sustich

6. Course Plan:

Module No.	Lecture Session	Reference	Learning outcomes
1: Introduction to Nanotechnology in the context of Renewable energy and the environment	L 1: Introduction to nanotechnology and classification of nanomaterials	TB 1 + Notes in Class	○ Describe the importance of nano-materials and the origin of their unique properties. ○ List the techniques used for synthesizing nanomaterials. ○ Summarize the concepts of energy harvesting and storage.
	L 2-3: Unique properties of nanomaterials, structural, electronic, magnetic, optical, and mechanical.		
	L 4-5: Techniques for nanomaterials synthesis, Nanocoating		
	L 6: Selection of nanomaterials for energy harvesting and storage		
2: Nanotechnology in the Renewable Energy Application	L 7-11: Fabrication, working principle, characterization, and nanotechnology applications of solar cells (i.e., organic, quantum dot, perovskite, and dye sensitized solar cells)	TB 2 + Notes in Class	○ Summarize the fabrication characterization techniques and working principles of solar cells, fuel cells, batteries, and CNT-based energy storage devices. ○ List the merits and motivations for using nanomaterials in the renewable energy and environmental sector ○ Solve elementary problems on cyclic voltammetry and impedance spectroscopy. ○ Application of nanomaterials in hydrogen production and its storage by different routes.
	L 12-14: Fabrication, working principle, characterization, and nanotechnology applications in fuel cells.		
	L 15-17: Fabrication, working principle, characterization and nanotechnology applications in batteries.		
	L 18-20: Synthesis and properties of carbon nanotubes and their application in fabricating energy storage devices		
	L 21-23: Nanotechnology for hydrogen production and storage		



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3: Nanotechnology in Environmental application	L24- 26: Introduction, Environmental fate & transport of nanomaterials	TB 3 +Journal papers + Class notes	• Explain the application of nanomaterials for environmental pollution (air, water, and solid) control. • Explain the mechanism of interaction between nanomaterials and the pollutants. • Explain the property-nanostructure correlation of the nanomaterials and their effects on pollutant removal. • Define the toxic effects of nanomaterials on the environment, relative remediation, risk assessment, and precautions.
	L27-31: Nanomaterials for wastewater treatment: reactivity, fate & lifetime, Delivery & transport issues. Nanomaterials as adsorbents: Adsorption at oxide nanoparticles, Solution interface, nanomaterial adsorbent for water and wastewater treatment.		
	L32-35: Nanomaterials for air pollution and waste management		
	L36-40: Toxicity of nanomaterials, Ecotoxicological impacts of nanomaterials, Paradigms in nanomaterial toxicity. Provide awareness about the social and economic impact of nanotechnology and to handle the techniques effectively, including nanomaterials risk assessment.		

7. Evaluation Scheme:

Component	Duration (min)	Weightage (%)	Date & Time	Nature of component (Close Book/ Open Book)
Mid-Semester Test	90	30	06/10 AN1	CB and/or OB
Comprehensive Exam	180	40	07/12 FN	CB and/or OB
Quizzes	30	10		Best 2 class tests out of 3 (CB)
Project Work		20		

8. Chamber Consultation Hour: Thursday (5:30 PM-6:30 PM)

9. Notices: The notices will be displayed/circulated on the Nalanda LMS and/or via emails.

10. Make-up Policy: Make-up will be granted for genuine cases only.

Instructor-in-Charge
Course No. BITS F429



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